INCH-POUND

MIL-M-28787/55B(SH)
AMENDMENT 3
8 March 1991
SUPERSEDING
AMENDMENT 2
17 April 1985

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

DRIVER, DIFFERENTIAL LINE, DIGITAL, KEY CODE KDM

This amendment forms a part of Military Specification MIL-M-28787/55B(EC), dated 12 January 1983, and is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense

PAGES 4 AND 5

TABLE I, delete and substitute new table I as printed on pages 3 and 4 of this amendment.

PAGE 6

* 4.4.2 Delete and substitute the following:

"4.4.2 Order of testing. Test conditions A through W shall be performed in any order with the following exceptions. Test conditions A, B, C, E, F, and L are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, B, C, E, F, or L are part of a sequence of tests, the order of testing shall be as follows: C, A, B, E, F, and L where applicable, followed by the remaining tests in the sequence."

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* 4 5 1: Delete and substitute the following:

"4.5 1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through W in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient When electrical testing is called for at the end of each temperature stabilization period, test conditions G through K and M through W shall be performed in the order specified in 4 4.2 using the initial limits for the applicable operating temperature At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2."

* 4 5 3: Delete and substitute the following:

"4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests G through K and M through W shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C electrical tests G through K and M through W shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

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TABLE XII, Test conditions K1 and K2, Max column: Delete "90" (two places) and substitute "120" (two places).

TABLE XII, Test conditions K3 through K10, Max column: Delete "30" and substitute "40"

TABLE XII, Test conditions K11 through K18, Max column: Delete "60" and substitute "80".

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TABLE XIV, Test conditions M1 and M2, Max column: Delete "244" (two places) and substitute "250" (two places).

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TABLES XVIII and XIX, delete and substitute new tables XVIII and XIX as printed on page 5 of this amendment

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FIGURE 21, delete and substitute new figure 21 as printed on page 6 of this amendment

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TABLE I. Module electrical requirements

Test condition	Parameter	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0 ± 5°C		60 ± 5°C		-55 ± 5°C		85 ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1-A20	V_{OUT} (low loaded)	III	S	10	800	10	800	10	800	10	800	10	800	mV
B1,B2	Input breakdown current	IV	S	0	225	0	300	0	300	0	300	0	300	uA
B3-B10	Input breakdown current	IV	S	0	75	0	100	0	100	0	100	0	100	uA
B11-B18	Input breakdown current	IV	S	0	150	0	200	0	200	0	200	0	200	uA
C	Isolation													
	Initial	V	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	V	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
D1,D2	Input capacitance	VI	Q	2	20	2	20	2	20	2	20	2	20	pF
D3-D18	Input capacitance	VI	Q	2	15	2	15	2	15	2	15	2	15	pF
E1-E18	Input clamp diode voltage	VII	S	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
F1-F20	Output clamp diode voltage	VIII	Q	0.3	1.5	0 3	1.5	0.3	1.5	0.3	1.5	0.3	1.5	V
G1-G28	Output voltage, low level	IX	X	0	400	0	400	0	400	0	400	0	400	mV
H1-H28	Output voltage, high level	X	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
J1-J18	Input current, low level	XI	X	0.40	1.73	0.40	1.73	0.40	1.73	0.40	1.73	0.40	1.73	mA
K1,K2	Input current, high level	XII	X	0	120	0	120	0	120	0	120	0	120	uA
K3-K10	Input current, high level	XII	X	0	40	0	40	0	40	0	40	0	40	uA
K11-K18	Input current, high level	XII	X	0	80	0	80	0	80	0	80	0	80	uA
L1-L20	Output short circuit current	XIII	S	40	120	40	120	40	120	40	120	30	120	mA

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0 ± 5°C		60 ± 5°C		-55 ± 5°C		85 ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
M1	Power supply current, low level	XIV	S	1	250	1	250	1	250	1	250	1	250	mA
M2	Power supply current, high level	XIV	S	1	250	1	250	1	250	1	250	1	250	mA
N1-N20	Transition time (t_{TLH})	XV	S	4	45	4	50	4	50	4	50	4	50	ns
P1-P20	Transition time (t_{THL})	XV	S	2	20	2	25	2	25	2	25	2	25	ns
R1-R20	Propagation delay time (t_{PLH})	XV	S	5	30	5	45	5	45	5	45	5	45	ns
S1-S20	Propagation delay time (t_{PHL})	XV	S	5	30	5	35	5	35	5	35	5	35	ns
T1-T20	Output crosstalk, low level	XVI	Q	0	1	0	1	0	1	0	1	0	1	V
U1-U20	Output crosstalk, high level	XVII	Q	0	750	0	750	0	750	0	750	0	750	mV
V	Power supply transient amplitude	XVIII	Q	0.0	1.3	0.0	1.3	0.0	1.3	0.0	1.3	0.0	1.3	V
W	Power supply capacitance	XIX	S	6.6	12.2	6.6	12.2	6.6	12.2	6.6	12.2	6.6	12.2	uF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified, limits are initial and end-of-life.

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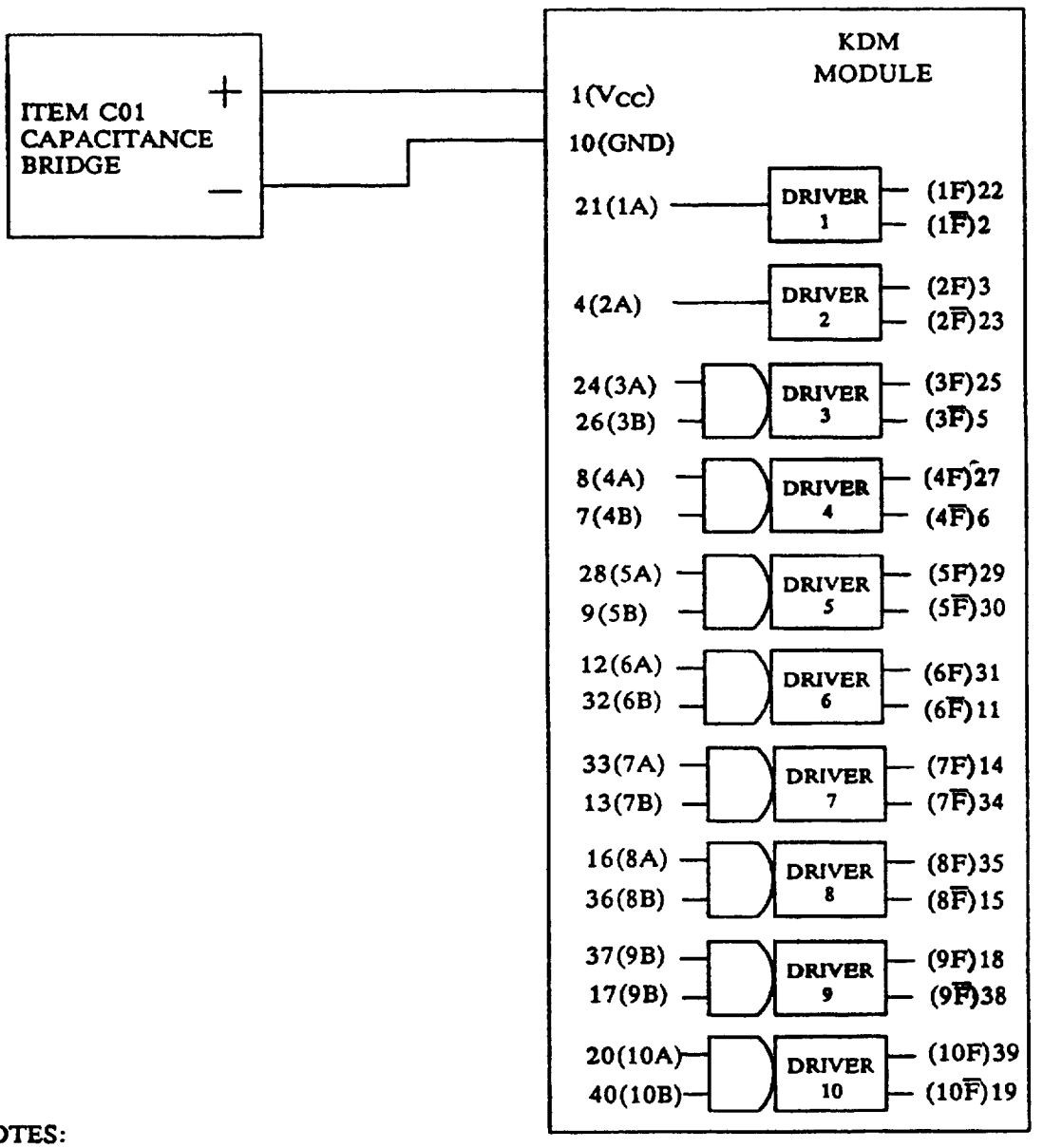
TABLE XVIII. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25 ± 5°C, ambient		Unit	Test figure
		Min	Max		
V	Power supply transient amplitude	0.0	1.3	V	19, 20

TABLE XIX. Power supply capacitance test.

Test condition	Parameter	Initial limits 25 ± 5°C, ambient		Unit	Test figure
		Min	Max		
W	Power supply capacitance	6.6	12.2	uF	21

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NOTES:
1. Null out the effect of the test leads prior to connecting to the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings
3 Item C01 setup.
   a Frequency = 1 kHz.
   b Vpk = 0 3 V maximum.

FIGURE 21 Power supply capacitance test circuit for table XIX

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FIGURE 22, delete in its entirety.

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